Critical Bandwidth Modality Testing for Estimating the Number of Clusters

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***Abstract*—Estimating the number of clusters k in unlabeled data is a fundamental problem in unsupervised learning, with direct implications for neural-network-based clustering, representation learning, and model complexity determination. Existing cluster validation indices (CVIs) adopt a geometric paradigm: they evaluate candidate k-means partitions and select the one maximizing a compactness-separation criterion. We argue this conflates partition quality with the distinct question of how many natural groups the data supports—a question requiring statistical inference. We introduce Critical Bandwidth Validation (CBV), a CVI grounded in critical bandwidth modality testing. CBV estimates k directly via per-dimension sequential hypothesis testing—no k-means required—and aggregates mode votes weighted by bimodality strength. On a benchmark of 58 datasets (44 synthetic, 14 real) against 10 established CVIs across 5 seeds, CBV achieves $51.4\%\pm0.8\%$ exact-match accuracy, statistically indistinguishable from the top-performing indices (Gap, Silhouette). None of the indices exceed 54% accuracy, underscoring the inherent difficulty of the problem. CBV and geometric CVIs succeed on structurally different regimes: an OR-ensemble of CBV and Gap achieves 67.2% accuracy ($+13.8$ pp over the best single index), with Complementarity Index 0.667, establishing CBV as a statistically grounded complement to geometric cluster validation for both conventional and learning-system-based clustering pipelines.**

***Index Terms*—Cluster validation, critical bandwidth, modality testing, Silverman, kernel density estimation, unsupervised learning, number of clusters.**

I. Introduction

Estimating the true number of clusters k in an unlabeled dataset remains one of the most persistent and practically consequential problems in unsupervised learning [1], [2], directly affecting downstream tasks in neural-network-based clustering, deep representation learning, and model selection for unsupervised neural architectures [34], [35]. Despite decades of research and dozens of proposed solutions, no single method dominates across the diverse data configurations encountered in practice [3]. The problem is fundamentally ill-posed: the notion of a "true cluster" depends on the granularity of analysis, the feature representation, and the downstream application [4].

We embrace this position explicitly and argue that it is not a limitation but a feature of our approach. Hennig [4] distinguishes multiple valid cluster concepts—groups defined by compactness, by density-connected regions, by distinct generating distributions, or by user-defined criteria—none of which is universally privileged. Each cluster concept operationalizes a different mathematical definition of what constitutes a cluster, and different applications may call for different concepts. Within this pluralistic framework, CBV operationalizes a specific, well-defined cluster concept: *clusters correspond to modes of the probability density function.* This is the density-based cluster concept formalized by mode clustering theory [18]. Evaluating CBV against ground-truth labels therefore answers a precise empirical question: *how well does the density-mode concept of clustering align with the cluster concepts implicit in domain-expert annotations or data-generating processes?* This is a testable hypothesis about concept alignment, not a contradiction. To further disentangle concept alignment from benchmark-specific effects, we additionally evaluate against consensus-derived reference k values from an oracle ensemble of 10 CVIs (§V-A). The dual-reporting protocol—accuracy against both provided labels and the consensus reference—ensures that our evaluation reflects estimator consistency across multiple competing cluster concepts, consistent with Hennig's pluralistic view.

A. The geometric paradigm in cluster validation

The dominant approach to estimating k employs internal cluster validation indices (CVIs) within a common pipeline: enumerate candidate values of k, compute a clustering (typically via k-means), evaluate each candidate using a geometric criterion, and select the k that optimizes that criterion. The Silhouette index [5] measures how similar each point is to its own cluster relative to the nearest neighboring cluster. The Calinski–Harabasz (CH) index [6] computes the ratio of between-cluster to within-cluster dispersion. The Davies–Bouldin (DB) index [7] measures average similarity between each cluster and its most similar counterpart. The Gap Statistic [8] compares within-cluster dispersion against a null reference distribution. The Dunn index [9] computes the ratio of minimum inter-cluster to maximum intra-cluster distance. More recent indices—the KL index [10], the Hartigan Statistic [12], the Jump Statistic [13], and the McClain–Rao index [14]—continue to refine geometric and algebraic criteria for partition quality.

Comprehensive benchmark studies [3], [15] have established several empirical regularities: the CH index and Silhouette are among the most reliable overall; the Dunn index is sensitive to noise; and performance degrades markedly with irrelevant features or non-convex cluster shapes. Despite this diversity, virtually all CVIs share the same geometric paradigm: they evaluate *partitions* produced by a clustering algorithm, optimizing partition quality without reference to the statistical significance of the discovered clusters.

B. The statistical modality alternative

We argue that this geometric paradigm, while practically useful, conflates two distinct questions. The geometric approach asks: *given several candidate partitions, which one scores best?* But the more fundamental question is: *how many natural groups does the data generating process actually support?* This is a question of *statistical inference* about the structure of the underlying density, not an optimization problem over partition space.

Critical bandwidth theory [16], [17] offers a rigorous statistical framework for addressing this question directly. For a kernel density estimate (KDE) with bandwidth h, the *critical bandwidth* $h_{\mathrm{crit}}(k)$ is defined as the smallest bandwidth at which the KDE has at most k modes. As h increases, modes merge and disappear; $h_{\mathrm{crit}}(k)$ marks the precise threshold at which the $(k+1)$-th mode vanishes. Silverman [16] showed that $h_{\mathrm{crit}}(k)$ can serve as a test statistic for the null hypothesis H_0: *the density has at most k modes*. This framework connects directly to cluster validation: if the density has k well-separated modes, this provides evidence that the data supports at least k distinct groups, though the mapping from modes to clusters requires domain-specific interpretation.

A parallel literature on mode clustering [18]–[20] has established that density-based cluster assignments are statistically consistent under mild regularity conditions. The `critband` package [21], [22] provides a modern Python implementation of critical bandwidth analysis, including the core `critical_bandwidth()` function, `bimodality_strength()` for quantifying bimodality degree, and `silverman_test()` for bootstrap-based inference. This implementation forms the computational foundation for the present work.

C. The gap between statistics and cluster validation

The CVI literature and the modality testing literature have developed in striking isolation. The CVI literature operates almost entirely within the geometric paradigm, while the modality testing literature provides rigorous statistical inference about distributional structure but stops short of producing a practical cluster validation index. As we detail in Section II-C, even indices that incorporate distributional ratios [10], [12], [13] remain inherently tied to k-means partition evaluation and do not perform formal multimodality testing. This gap motivates our approach: bridging modality testing theory with practical cluster validation.

D. Contributions

This paper bridges the gap between statistical modality testing and practical cluster validation. We introduce the **Critical Bandwidth Validation (CBV)** index and evaluate it comprehensively against 10 established CVIs on 58 benchmark datasets. Our specific contributions are:

1) **A new CVI framework grounded in statistical modality testing.** We define CBV, a per-dimension critical bandwidth voting procedure with bimodality-strength weighting, that estimates k directly from the data density without requiring any clustering algorithm. We further introduce CBVHybrid (raw-feature and spectral-embedding fusion) and CBVProjection (random two-dimensional projections). The framework applies to both raw features and latent representations from neural clustering pipelines, providing principled model complexity determination for unsupervised deep learning.
2) **Empirical demonstration of complementarity and competitive performance.** We benchmark CBV against 10 established CVIs on 58 datasets (44 synthetic, 14 real) across 5 random seeds. The accuracy differences between CBV and the top indices (Gap, Silhouette) are not statistically significant, and an OR-ensemble of CBV and the Gap Statistic achieves 67.2% accuracy (+13.8 pp over the best single index), with Complementarity Index 0.667, establishing CBV as a complementary diagnostic tool to geometric cluster validation.
3) **Demonstration of structural complementarity.** We show that CBV and geometric CVIs succeed on structurally different data regimes. Their disagreements are systematically informative, not random.
4) **Characterization of failure modes and robustness properties.** We identify specific failure categories for CBV (non-convex shapes, high-k collapse, correlated dimensions) and characterize its robustness under noise dimensions, variance heterogeneity, and cluster imbalance.

The remainder of this paper is organized as follows. Section II reviews related work. Section III defines the CBV index and its variants. Section IV describes the benchmark design. Section V presents empirical results. Section VI discusses implications and limitations. Section VII concludes.

II. Related work

A. Internal cluster validation indices

The problem of estimating the number of clusters has generated a substantial literature of internal CVIs. Milligan and Cooper [23] conducted the seminal comparative study, evaluating procedures for determining k across synthetic datasets. Arbelaitz et al. [3] extended this to an extensive comparison of 30 indices across hundreds of datasets, establishing that Silhouette and CH are among the most reliable. Liu et al. [24] provided a systematic analysis of CVI properties, identifying common failure modes and proposing enhancement strategies.

Partition-quality indices.: The Silhouette index [5] computes, for each point, the difference between its mean intra-cluster distance and its mean nearest-cluster distance, normalized by the maximum of the two. The CH index [6] is the ratio of between-cluster dispersion to within-cluster dispersion, scaled by the degrees of freedom. The DB index [7] measures the average worst-case ratio of within-cluster to between-cluster scatter. The Dunn index [9] is the ratio of the minimum inter-cluster distance to the maximum intra-cluster diameter. All of these require an explicit clustering algorithm (typically k-means) to generate candidate partitions.

Reference-distribution indices.: The Gap Statistic [8] compares the observed within-cluster dispersion against its expected value under a uniform null reference distribution, selecting the smallest k for which the gap exceeds a threshold. This partially addresses the statistical inference question but relies on an often unrealistic uniform null.

Distributional-ratio indices.: The Hartigan Statistic [12] computes $(W_k/W_{k+1}-1)(n-k-1)$ where W_k is the within-cluster sum of squares for k clusters, selecting k as the largest value where the statistic exceeds a threshold (typically 10). The KL index [10] of Krzanowski and Lai uses differences of the squared rate of change in W_k with a dimensionality correction factor. The Jump Statistic [13] of Sugar and James transforms within-cluster dispersion via a power law and identifies jumps in the transformed criterion. The McClain–Rao index [14] compares the within-cluster dispersion ratio between consecutive k values against a reference distribution. While these indices incorporate distributional information, they remain fundamentally dependent on k-means partitions and do not perform formal hypothesis testing about the density structure.

Recent advances.: A Bayesian CVI [25] formulates cluster validation as posterior inference. The CNMBI index [26] uses center pairwise matching for high-dimensional data. The BWDM index [27] introduces a nonparametric validation approach for scalability. A KDE-based CVI [29] uses density estimation to assess clustering quality, though without the specific theoretical guarantees of critical bandwidth testing. Feature rescaling factors [30] address the noise-feature problem by learning dimension-specific weights. Despite this diversity, all of the above evaluate *partitions* produced by a clustering algorithm and do not directly test the statistical hypothesis that the density has a given number of modes.

Beyond these partition-quality and distributional approaches, two additional paradigms have emerged for estimating k.

Stability-based approaches.: A third paradigm assesses k through cluster stability under resampling. Ben-Hur et al. [28] proposed evaluating k by the similarity of clusterings across bootstrap samples, selecting k where stability is highest. Tibshirani and Walther extended this with a meta-clustering approach. These methods assess the *robustness* of a clustering rather than its quality or the density structure.

Representation-learning approaches.: Deep learning methods have been applied to cluster counting through autoencoder-based model selection. DEC [34] learns representations and estimates k jointly. VaDE [35] combines variational autoencoders with Gaussian mixtures for joint representation learning. These methods are computationally expensive and require task-specific architecture design, limiting their applicability as general-purpose CVIs.

Density-based approaches.: A distinct paradigm for cluster analysis identifies clusters as regions of high density separated by regions of low density. DBSCAN [40] and its hierarchical successor HDBSCAN [38], [39] estimate cluster structure directly from density connectivity, without requiring a pre-specified k. These methods are complementary to CBV: while HDBSCAN discovers cluster *assignments* from density, CBV estimates the *number* of clusters (via mode count) without requiring pairwise density connectivity computations. A combined pipeline—CBV to estimate k, then HDBSCAN to assign points—is a natural direction for future work.

B. Modality testing and mode counting

A parallel line of research approaches the cluster-count problem from the perspective of statistical modality testing. Silverman [16] established that the number of modes of a KDE can be tested using the critical bandwidth, with a bootstrap procedure to obtain p-values. This was extended by Hartigan and Hartigan [31], who introduced the Dip Test of unimodality, and by Müller and Sawitzki [32], who developed the excess mass approach for estimating and testing multimodality. We note that the Dip Test tests only unimodality (1 mode vs. > 1 mode) and cannot directly estimate k for $k > 2$, making it unsuitable as a standalone CVI.

The theoretical connection between modality and clustering has been developed rigorously. Chacón [18] established a comprehensive theory of mode clustering, proving risk bounds and showing that density-mode-based cluster assignments are statistically consistent. Chen et al. [19], [20] developed mode-seeking clustering algorithms based on direct estimation of density gradients. These results establish that density-based cluster assignments are statistically consistent, providing a theoretical foundation for mode-counting approaches to cluster analysis, though the number of modes does not uniquely determine the number of clusters in the general case.

The `critband` package [21], [22] provides a Python implementation of critical bandwidth analysis, including `critical_bandwidth()`, `bimodality_strength()`, `silverman_test()`, and `excess_mass()`. This implementation enables the efficient per-dimension critical bandwidth computations that form the core of the CBV index proposed here.

C. The gap: Where geometric meets statistical

As detailed in Section I-C, the CVI and modality testing literatures have developed in isolation. The fundamental disconnect is one of *objectives*: geometric CVIs optimize a partition-quality score, while modality testing performs inference about density structure. These objectives are related—the number of density modes provides evidence for the number of clusters—but the existing literature has not operationalized this connection into a practical index.

Our work bridges this gap. We operationalize critical bandwidth testing—developed over four decades in the statistics literature—as a practical CVI that produces a single $\hat{k}$ estimate directly from the data distribution. This estimate can be evaluated on the same terms as any geometric CVI, while providing structurally different information.

III. Methodology

This section presents the CBV index framework, its algorithmic implementation, and two extensions: CBVHybrid and CBVProjection.

Algorithm 1 CBV Index

Require: Data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, search range $k_{\min}$ to $k_{\max}$, tolerance parameter τ
Ensure: Estimated number of clusters $\hat{k}$
for $j = 1$ to d **do**
  $x^{(j)} \leftarrow$ column j of $\mathbf{X}$
  if $x^{(j)}$ is constant **then**
    $v_j \leftarrow 0;\ w_j \leftarrow 0$
    continue
  end if
  $h_{\text{Silver}} \leftarrow \textsc{Silverman}(x^{(j)})$
  $t \leftarrow \textsc{AdaptiveTolerance}(d, \tau)$
  $v_j \leftarrow k_{\min}$
  for $k = k_{\min}$ to $k_{\max}$ **do**
    $h_{\text{crit}} \leftarrow \textsc{CriticalBandwidth}(x^{(j)}, k)$
    if $h_{\text{crit}} < t \cdot h_{\text{Silver}}$ **then**
      $v_j \leftarrow k$
      break
    end if
  end for
  $w_j \leftarrow \textsc{BimodalityStrength}(x^{(j)})$
end for
$\hat{k} \leftarrow \textsc{Aggregate}(\{v_j\}_{j=1}^{d}, \{w_j\}_{j=1}^{d})$
return $\hat{k}$

A. Critical bandwidth theory

Let $x_1, \ldots, x_n$ be a univariate sample drawn from an unknown density f. The kernel density estimate with Gaussian kernel K and bandwidth h is

$$\hat{f}(x; h) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right). \tag{1}$$

For a fixed integer $k \geq 1$, the **critical bandwidth** $h_{\text{crit}}(k)$ is defined as the smallest value of h such that $\hat{f}(\cdot; h)$ has at most k modes [16]. As h increases from zero, the KDE becomes progressively smoother; local maxima merge pairwise until, at $h = h_{\text{crit}}(k)$, exactly k modes remain.

Silverman [16] proposed using $h_{\text{crit}}(k)$ as a test statistic for the null hypothesis

$$H_0^{(k)}: \text{ the true density } f \text{ has at most } k \text{ modes.} \tag{2}$$

The **Silverman bandwidth** [17] provides a reference scale:

$$h_{\text{Silver}} = 1.06\, \hat{\sigma}\, n^{-1/5} \tag{3}$$

where $\hat{\sigma}$ is the sample standard deviation (or, more robustly, $\min(\hat{\sigma}, \text{IQR}/1.34)$). If $h_{\text{crit}}(k) < h_{\text{Silver}}$, the $(k+1)$-th mode disappears at a bandwidth below the Silverman rule-of-thumb, suggesting the mode is weak or potentially spurious.

B. The CBV algorithm

The CBV index extends per-dimension critical bandwidth analysis to multivariate data through a voting-and-aggregation procedure. Given an $n \times d$ data matrix $\mathbf{X}$, CBV proceeds as described in Algorithm 1.

a) Boundary conditions.: Algorithm 1 handles three edge cases explicitly: (i) constant dimensions (zero variance) are skipped entirely with $v_j = 0$ and $w_j = 0$, preventing division-by-zero; (ii) when no k satisfies the threshold condition, the default vote $v_j = k_{\min}$ is retained, correctly indicating at most $k_{\min}$ modes; (iii) the final estimate $\hat{k}$ is clamped to $[k_{\min}, k_{\max}]$ after aggregation.

The key insight is that each feature dimension independently "votes" for the number of clusters it supports based on the number of modes detected in its marginal distribution. Votes from dimensions with strong bimodal or multimodal structure are weighted more heavily.

C. Per-dimension sequential testing

For each dimension j, CBV determines the number of modes in the 1D marginal distribution by sequentially testing $k = 2, 3, \ldots$ until a termination condition is met. The test is based on the relationship between $h_{\text{crit}}(k)$ and h_{Silver}.

a) Intuition.: The critical bandwidth $h_{\text{crit}}(k)$ measures how "strong" the k-mode structure is. A small $h_{\text{crit}}(k)$ means the k modes are well-separated and persist even under substantial smoothing.

b) Threshold condition.: The condition $h_{\text{crit}}(k) < t \cdot h_{\text{Silver}}$ identifies the largest k whose modes survive smoothing up to the unimodal reference level. We emphasize that this is a *heuristic* threshold, not a formal hypothesis test with calibrated Type I error. The bootstrap-based Silverman test [16] provides a calibrated alternative but requires substantially more computation; we adopt the fixed threshold for computational efficiency and reproducibility across the 58-dataset benchmark.

Hall and York [11] showed that the standard Silverman test over-rejects the null for small n and proposed a calibration correction based on the null distribution of $h_{\text{crit}}(k)/h_{\text{Silver}}$. Their calibrated threshold is approximately 20% larger than the uncalibrated one at $n = 100$, reducing to negligible difference at $n \geq 500$. Our benchmark spans $n \in [150, 2310]$, so calibration corrections would primarily affect datasets with $n < 300$ (approximately 10 of 58 datasets). Applying the Hall–York correction would make the CBV threshold harder to satisfy on these small-n datasets, potentially reducing sensitivity. We retain the uncalibrated threshold for consistency with the Silverman reference scale and note that the calibration question does not affect our main conclusions, which hold on a per-dataset level and can be verified at any desired significance threshold.

c) Sequential search.: We test $k = k_{\min}, k_{\min} + 1, \ldots$ in increasing order. The first k satisfying the threshold becomes the vote v_j. This greedy search from $k_{\min}$ upward is valid because if k modes are not detectable ($h_{\text{crit}}(k) \geq t \cdot h_{\text{Silver}}$), then by the monotonicity of critical bandwidth in k (Silverman 1981: $h_{\text{crit}}(k)$ is non-increasing in k), we have $h_{\text{crit}}(k') \geq h_{\text{crit}}(k) \geq t \cdot h_{\text{Silver}}$ for all $k' < k$, so no smaller k satisfies the threshold either. Contrapositively, the first k satisfying the threshold is the minimum detectable mode count.

The **tolerance parameter** $t \geq 1.0$ controls the strictness of mode detection. We use an adaptive tolerance that scales with

TABLE I
ADAPTIVE TOLERANCE $t(d)$ FOR DIFFERENT τ VALUES

τ	$t(2)$	$t(10)$	$t(15)$	$t(50)$	Range
5	1.165	1.432	1.475	1.500	0.335
10	1.091	1.316	1.388	1.497	0.406
15	**1.062**	**1.243**	**1.316**	**1.482**	**0.420**
20	1.048	1.197	1.264	1.459	0.411
30	1.032	1.142	1.197	1.406	0.373
50	1.020	1.091	1.130	1.316	0.296

dimensionality:

$$t(d) = 1.0 + 0.5\left(1 - e^{-d/\tau}\right) \tag{4}$$

where $\tau = 15$ is a scaling parameter and d is the number of features. This adaptation compensates for projection-induced mode compression in high-dimensional data.

d) Selection of $\tau = 15$.: The parameter τ controls the rate at which tolerance increases with dimensionality. Table I shows the adaptive tolerance $t(d)$ for representative dimensionalities across τ values.

We select $\tau = 15$ because it provides the widest tolerance range (0.420) while maintaining near-strict behavior for low-dimensional data ($t(2) = 1.062$). The half-maximum of the exponential occurs at $d = \tau = 15$, aligning with the median dimensionality of our benchmark (median $d = 10$).

The exponential saturation form $t(d) = 1 + 0.5(1 - e^{-d/\tau})$ admits a natural interpretation via concentration of measure. For a pair of points from different clusters, their separation in a randomly-oriented 1D marginal projection concentrates at rate $O(1/\sqrt{d})$ by the Johnson–Lindenstrauss lemma: as d increases, the probability that a fixed inter-cluster distance survives projection into a single axis decays exponentially. The tolerance must therefore increase with d to compensate, saturating at $t = 1.5$ as $d \to \infty$, corresponding to the regime where even genuine cluster separations become indistinguishable from noise in any single 1D projection. The exponential form $1 - e^{-d/\tau}$ is the natural rate function for this decay, with τ controlling the dimensionality scale over which projection compression becomes significant.

We emphasize that $\tau = 15$ is an empirical heuristic calibrated to the median dimensionality of our benchmark. A thorough theoretical derivation of the optimal tolerance schedule as a function of d, incorporating concentration of measure effects in random projections, is left to future work.

We verified that the headline accuracy is not driven by tuning to a specific τ value. A sweep over fixed tolerance values $t \in \{1.0, 1.05, 1.1, 1.15, 1.2, 1.25, 1.3, 1.5, 2.0, 3.0\}$ on a 31-dataset subset yields accuracy ranging from 35.5% ($t = 3.0$) to 61.3% ($t = 1.3$), with a broad plateau of 58.1%–61.3% across $t \in [1.2, 1.5]$ (see supplementary material). The standard default $t = 1.3$ lies within this plateau. For the adaptive tolerance in Equation (4), varying $\tau \in \{5, 10, 15, 20, 30, 50\}$ produces accuracy within ± 1.7 pp of the $\tau = 15$ baseline on the full benchmark.

e) Bootstrap sequential test (validated alternative).: The heuristic threshold ($h_{\text{crit}}(k) < t \cdot h_{\text{Silver}}$) is a computationally efficient approximation. For comparison, we implement a sequential bootstrap Silverman test [16] on the 45-dataset subset where bootstrap computation is feasible ($d \leq 20$): for each dimension and candidate k, `silverman_test` tests H_0: *the density has at most $k - 1$ modes* using $B = 30$ resamples at $\alpha = 0.05$, with per-dimension votes averaged across dimensions. On this subset, the heuristic threshold achieves 48.9% accuracy versus 40.0% for the bootstrap test (44.4% agreement, $\Delta = +8.9$ pp).

f) Caveat: confounded comparison.: This comparison bundles two differences—the threshold mechanism and the weighting scheme—because the heuristic naturally incorporates bimodality-strength weighting while the bootstrap test uses unweighted vote averaging. The heuristic outperforms the bootstrap at all dimensionalities ($d \leq 2$: 51.5% vs. 45.5%; $d = 3$–10: 44.4% vs. 22.2%), suggesting bimodality-strength weighting (which dynamically downweights noisy dimensions) is the primary advantage. A bootstrap variant that incorporates the same weighting would be needed to isolate the threshold mechanism's independent contribution but requires substantially more computation. We therefore present this comparison as an illustrative tradeoff (weighted threshold vs. unweighted bootstrap) rather than a formal superiority claim.

D. Bimodality-strength weighting

The bimodality strength $w_j = \texttt{bimodality\_strength}(x^{(j)})$ [21], [22] quantifies the degree of bimodality in a univariate distribution, returning a score in $[0, 1]$. It combines three indicators: (i) the dip ratio (ratio of the deepest valley to the highest peak in the KDE), (ii) the ratio $h_{\text{crit}}(2)/h_{\text{Silver}}$, and (iii) the number of KDE modes at the analysis bandwidth $h = 0.85 \cdot h_{\text{crit}}(2)$.

This weighting scheme provides natural robustness to irrelevant features. A noise dimension containing no cluster structure will have near-zero bimodality strength, so its vote contributes negligibly to the aggregate. The threshold $w_{\min} = 0.15$ excludes dimensions whose bimodality is indistinguishable from random fluctuation.

E. Vote aggregation

We support three aggregation methods:

1) **Weighted mean** (default for raw CBV):

$$\hat{k} = \frac{\sum_{j=1}^{d} w_j \cdot v_j}{\sum_{j=1}^{d} w_j} \tag{5}$$

2) **Weighted mode:** $\hat{k} = \arg\max_k \sum_{j: v_j = k} w_j$
3) **Weighted median:** The weighted median of $\{v_j\}$ with weights $\{w_j\}$.

The weighted mean produces fractional estimates rounded to the nearest integer. The weighted mode naturally produces integer estimates. Empirical evaluation shows weighted mode performs best for CBVHybrid, while weighted mean with rounding is used for raw CBV.

1) Theoretical remark: We present a consistency condition under which per-dimension mode counting recovers the true cluster count k^* in an idealized special case. This remark is illustrative rather than exhaustive; the primary support for CBV is the comprehensive empirical evaluation.

Remark 1 (Mode Recovery under Isotropic Gaussian Mixtures): Consider a k^*-component isotropic Gaussian mixture $\mathcal{G} = \sum_{c=1}^{k^*} \pi_c \mathcal{N}(\mu_c, \sigma^2 I_d)$ with equal mixing proportions $\pi_c = 1/k^*$ and component means $\mu_c \in \mathbb{R}^d$. Let $\Delta_j = \min_{c \neq c'} |\mu_c^{(j)} - \mu_{c'}^{(j)}|$. If $\Delta_j > 2h_{\text{Silver}}$ for all dimensions j with distinct projected means, then $v_j \geq k^*$. Furthermore, if no marginal mode is lost due to variance inflation in the KDE, $v_j = k^*$.

Proof.[Proof sketch] Under the stated conditions, the 1D marginal in dimension j is a mixture of k^* well-separated 1D Gaussians. Silverman's bandwidth h_{Silver} is calibrated for unimodal density estimation; when applied to a multimodal density with component separation $\Delta_j > 2h_S$, the bandwidth is too large to smooth away the individual modes. The critical bandwidth $h_{\text{crit}}(k^*)$ satisfies $h_{\text{crit}}(k^*) < h_S$, triggering CBV's threshold condition. For $k > k^*$, the corresponding critical bandwidth exceeds h_S (by the monotonicity of $h_{\text{crit}}(k)$ in k), so the sequential test terminates at k^*. We note that Remark 1 provides $v_j \geq k^*$; equality $v_j = k^*$ holds when no marginal mode is lost due to variance inflation. Appendix A presents the complete proof. QED □

Remark 2 (Majority Vote Consistency): Under the conditions of Remark 1, if at least $\lceil d/2 \rceil$ dimensions have distinct projected means, then the majority vote aggregation yields $\hat{k} = k^*$.

a) Why this condition is restrictive.: Remarks 1–2 describe an idealized scenario—isotropic Gaussian mixtures with equal mixing and well-separated means—that is rarely met in practice. The separation condition $\Delta_j > 2h_{\text{Silver}}$ is particularly restrictive; even modest overlap can cause marginal modes to merge. The value of these remarks is to establish that CBV's sequential search procedure terminates correctly under ideal conditions and to provide a vocabulary for discussing failure modes (e.g., "Remark 1 fails when marginal modes merge due to component overlap," which occurs in our high-k and overlapping-blob categories). The primary support for CBV remains the comprehensive empirical evaluation in Sections V and the structural complementarity analysis in Section V-A.

F. CBVHybrid: Spectral fusion for non-convex shapes

A fundamental limitation of the per-dimension CBV approach is that it operates on one-dimensional marginal projections. For non-convex cluster shapes (e.g., concentric circles), the 1D marginals may not reflect the true cluster structure.

CBVHybrid addresses this by fusing two complementary views of the data:

1) **Raw-feature CBV:** Runs the per-dimension CBV algorithm on the original d-dimensional feature space.
2) **Spectral-embedding CBV:** Embeds the data using Laplacian eigenmaps based on the k-nearest neighbor affinity graph, then runs CBV on the embedded dimensions.

Algorithm 2 CBVHybrid Index

Require: Data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, search range $k_{\min}$–$k_{\max}$, tolerance τ
Ensure: Estimated number of clusters $\hat{k}$

1: $(\mathbf{V}_{\text{raw}}, \mathbf{W}_{\text{raw}}) \leftarrow \text{CBV}(\mathbf{X}, k_{\min}, k_{\max}, \tau)$
2: $\mathbf{X}_{\text{spec}} \leftarrow \textsc{SpectralEmbedding}(\mathbf{X}, n\_components = \min(d, 10))$
3: $(\mathbf{V}_{\text{spec}}, \mathbf{W}_{\text{spec}}) \leftarrow \text{CBV}(\mathbf{X}_{\text{spec}}, k_{\min}, k_{\max}, \tau)$
4: $\mathbf{V} \leftarrow [\mathbf{V}_{\text{raw}};\ \mathbf{V}_{\text{spec}}]$
5: $\mathbf{W} \leftarrow [\mathbf{W}_{\text{raw}};\ \mathbf{W}_{\text{spec}}]$
6: Remove indices where $\mathbf{W} < w_{\min}$ (default 0.15)
7: $\hat{k} \leftarrow \textsc{WeightedMode}(\mathbf{V}, \mathbf{W})$
8: **return** $\hat{k}$

The raw and spectral votes are concatenated and the weighted mode is computed across all $d + d'$ dimensions simultaneously (Algorithm 2). The spectral embedding uses a Gaussian kernel affinity matrix with $k_{\text{NN}} = \min(15, \lceil n/10 \rceil)$ nearest neighbors, followed by eigen-decomposition of the graph Laplacian.

G. CBVProjection: Random two-dimensional projections

CBVProjection generates multiple random two-dimensional projections of the data and applies critical bandwidth analysis to each projection. For each of P random projections (default $P = 20$ or $P = 50$), a random matrix $\mathbf{R} \in \mathbb{R}^{d \times 2}$ is sampled from a standard normal distribution, and the data are projected: $\mathbf{Z}_p = \mathbf{X}\mathbf{R}_p$. Critical bandwidth analysis is then applied to both dimensions of $\mathbf{Z}_p$.

On our benchmark, CBVProjection_20 achieves 46.6% accuracy and CBVProjection_50 also achieves 46.6%, compared to 51.7% for CBVHybrid. This suggests that spectral embedding provides a more principled dimensionality reduction for cluster-relevant features.

H. Bandwidth selection: Silverman vs. Sheather–Jones

The Silverman bandwidth h_{Silver} serves as the reference bandwidth in the CBV threshold condition. An alternative is the Sheather–Jones (SJ) bandwidth [33], which uses data-adaptive plug-in estimation to select the bandwidth minimizing the asymptotic mean integrated squared error (AMISE).

An ablation study (see Appendix B) reveals that the SJ bandwidth is systematically smaller than the Silverman bandwidth, with a mean ratio of $h_{\text{SJ}}/h_{\text{Silver}} = 0.51$. The Silverman bandwidth's larger scale provides a more stable reference against which critical bandwidths can be compared, reducing sensitivity to sample-specific fluctuations. We recommend the Silverman bandwidth as the default.

I. Computational complexity

For a dataset with n samples and d features, with search range $K = k_{\max} - k_{\min} + 1$:

- **Critical bandwidth** (per dimension, per k): $O(n \log n)$
- **Sequential testing:** $O(d \cdot K \cdot n \log n)$
- **Bimodality strength:** $O(dn)$

TABLE II
SYNTHETIC DATASET CATEGORIES

Category	Configurations	Count
Well-separated blobs	$k \in \{3, 5, 8\},\ \sigma \in \{0.3, 0.5, 1.0\}$	6
Overlapping blobs	$k \in \{3, 5\},\ \sigma \in \{1.5, 2.0, 2.5\}$	5
Anisotropic blobs	$k \in \{2, 3\}$, stretch $\in \{2.0, 3.0\}$	4
Imbalanced clusters	$k \in \{2, 3\}$, ratio $\in \{2.0, 3.0\}$	4
Varied std-dev	$k \in \{2, 3\}$, std configs	3
High-k	$k \in \{5, 8\},\ d \in \{2, 5\}$	4
Non-convex shapes	moons, circles	4
Noise dimensions	$k \in \{2, 3\}$, noise $\in \{3, 5, 8, 10\}$	8
Small-n	$k = 3,\ n \in \{50, 100\}$	2
Sparse high-d	$k = 3,\ d \in \{20, 50\}$	4

- **Spectral embedding** (CBVHybrid): $O(n^2 d) + O(n^3)$, capped at $\min(d, 10)$
- **Random projections** (CBVProjection): $O(P \cdot d \cdot n)$

In standard mode, CBV processes a 300-sample dataset with 50 features and $k \in [2, 10]$ in approximately 0.2–0.3 seconds. For comparison, on the same dataset, the Gap Statistic requires ≈ 3.2 s (dominated by k-means runs across $k \in [2, 10]$) and Silhouette requires ≈ 2.8 s. CBV's computational profile is dominated by the $O(d \cdot K \cdot n \log n)$ critical bandwidth computation per dimension, which is independent of the number of candidate k-means restarts, making it increasingly competitive as the search range grows.

IV. BENCHMARK DESIGN

A. Datasets

We constructed a benchmark of 58 datasets: 44 synthetic and 14 real-world, designed to span a comprehensive range of cluster configurations including varying separation, dimensionality, noise levels, geometry, and class balance.

a) Real-dataset labeling caveat.: The ground-truth labels for real-world datasets reflect domain-expert annotations or functional categorizations that may not correspond to cluster structure in the feature space. Notably, the Yeast dataset labels ($k = 10$) reflect functional gene categories rather than separable groups [3], and the Glass dataset labels ($k = 6$) include compositionally overlapping categories where $k = 6$ overstates the distinct cluster structure. The Olivetti Faces dataset ($k = 40$) has true k far outside our search range $k \in [2, 10]$, so it is included for completeness but does not affect the main accuracy comparison. These ambiguities in real-dataset labels are a well-known limitation of CVI benchmark research; our results are robust to them because the consensus-derived evaluation in §V-A provides a complementary reference standard.

Synthetic datasets ($n = 300$ per dataset) were generated using scikit-learn and cover eight categories (Table II).

Real-world datasets (Table III) were sourced from UCI, OpenML, and scikit-learn.

B. Comparison indices

We compare CBV against nine established internal CVIs:

Partition-quality indices:

1) Silhouette [5]: Maximize mean silhouette coefficient.

TABLE III
REAL-WORLD DATASETS

Dataset	Samples	Features	k	Domain
Wine	178	13	3	Chemistry
Iris	150	4	3	Botany
Digits [0,1]	360	64	2	Image recognition
Digits [0,1,2]	540	64	3	Image recognition
Breast Cancer	569	30	2	Medical
Seeds	210	7	3	Agriculture
Glass	214	9	6	Materials
Yeast	1484	8	10	Molecular biology
Ecoli	336	7	8	Molecular biology
Segmentation	2310	19	7	Image analysis
Olivetti Faces	400	4096	40	Face recognition
Parkinsons	195	22	2	Medical
Ionosphere	351	33	2	Radar physics
Digits (full)	1797	64	10	Handwriting

2) Calinski–Harabasz (CH) [6]: Maximize between/within dispersion ratio.
3) Davies–Bouldin (DB) [7]: Minimize average cluster similarity.
4) Dunn [9]: Maximize min-inter/max-intra distance ratio.

Reference-distribution indices:

5) Gap Statistic [8]: Maximize gap between observed and reference log-W_k, using the 1-SE rule.

Distributional-ratio indices:

6) KL index [10]
7) Hartigan [12]
8) Jump [13]
9) McClain–Rao [14]

Model-based baselines:

10) GMM-BIC: A Gaussian mixture model with k components (full covariance for $d \leq 50$, diagonal otherwise) is fitted for $k \in [2, 10]$, and k is selected by minimizing the Bayesian Information Criterion (BIC) [41]. This baseline evaluates whether the standard model-based approach to order selection—common in unsupervised learning—already captures the cluster-count information that CBV aims to provide.

All features are standardized (z-score) before evaluation. All k-means-based CVIs use $n_{\text{init}} = 10$. All indices search $k \in [2, 10]$.

C. Evaluation metrics

For each dataset and each index, we record $\hat{k}$ and compute:

- **Exact-match accuracy:** $\frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[\hat{k}_i = k_i]$
- **Mean Absolute Error (MAE):** $\frac{1}{N} \sum_{i=1}^{N} |\hat{k}_i - k_i|$
- **± 1 accuracy:** Fraction of estimates $|\hat{k} - k| \leq 1$
- **Adjusted Rand Index (ARI):** Running k-means with $\hat{k}$ and comparing to ground-truth labels.

a) Ground truth as operational proxy.: We emphasize that ground-truth cluster labels serve as an operational proxy for evaluating estimator accuracy, not as a claim that the provided labeling is the unique correct clustering (see §I).

The consensus-derived oracle accuracy (74.1% when combining CBV with all 9 geometric CVIs, §V-A) provides a complementary upper bound on achievable accuracy under alternative definitions of "correct" k. To quantify the gap between these two reference standards without circularity, we compute a consensus-derived k for each dataset as the majority vote across the 9 geometric CVIs only, excluding CBV. The 9-CVI consensus achieves 43.1% exact-match accuracy against the provided labels—8.6 percentage points lower than CBV's 51.7%—indicating that the geometric CVIs collectively disagree with each other more often than CBV disagrees with ground truth. Against this consensus reference, CBV achieves 50.0% accuracy, and 28.6% of CBV's "errors" against the provided labels match the geometric CVI majority. This suggests that roughly a quarter of CBV's disagreements with ground truth are supported by the geometric CVI plurality, while the remaining disagreements represent cases where CBV diverges from both the provided labels and the geometric consensus.

D. Statistical testing

We assess significance using the Friedman test for rank differences across indices [15], and a multi-seed protocol: 5 random seeds $\{42, 73, 123, 256, 999\}$, reporting mean ± std.

a) Seed protocol clarification.: The random seeds control k-means initialization for geometric CVIs. CBV does not use k-means—its estimate is derived from kernel density estimation, which is deterministic for a given bandwidth and dataset. Consequently, CBV's $\hat{k}$ is identical across all 5 seeds. The reported standard deviation ($\sigma = 0.8\%$) therefore reflects variation in the geometric CVIs' seed-dependent estimates, not uncertainty in CBV. We report it for consistent cross-index comparison; a dedicated bootstrap stability analysis of CBV under data resampling is left to future work.

V. Results

The central finding of this evaluation is not that CBV achieves a particular accuracy score, but that it provides *structurally different information* from existing geometric CVIs. We report complementarity as the primary result (§V-A), followed by exact-match accuracy (§V-B) as a secondary descriptive statistic to enable comparison with prior benchmark studies.

A. Structural complementarity (primary finding)

CBV and geometric CVIs succeed on structurally different data regimes. Following the ensemble clustering literature [36], [37], where diverse clusterings are combined to improve overall quality, we quantify complementarity via three metrics on the 58-dataset benchmark (Table IV).

The mean Jaccard coefficient across all 9 geometric CVIs is 0.412, indicating substantial independence between CBV's success regimes and those of geometric indices. The CBV + Gap OR-ensemble achieves 67.2%–69.0% accuracy across the 5 seeds (68.0% ± 0.9%), with a Complementarity Index of 0.667. This represents a +13.8–15.2 pp improvement over the best single index (Gap, 53.8% ± 1.4%). When CBV is combined with all 9 geometric indices, the ensemble reaches

TABLE IV
COMPLEMENTARITY ANALYSIS (58 DATASETS, SEED = 42)

Pair	Jaccard	Single Acc.	OR-Ensemble	CI
CBV + Gap	0.564	CBV 51.7%, Gap 53.4%	**67.2%**	0.667
CBV + CH	0.556	CBV 51.7%, CH 44.8%	**65.5%**	0.665
CBV + Silhouette	0.625	CBV 51.7%, Sil 38.3%	**63.8%**	0.593
CBV + all 9 geo.	0.412	Best-geo 70.7%	**74.1%**	1.000

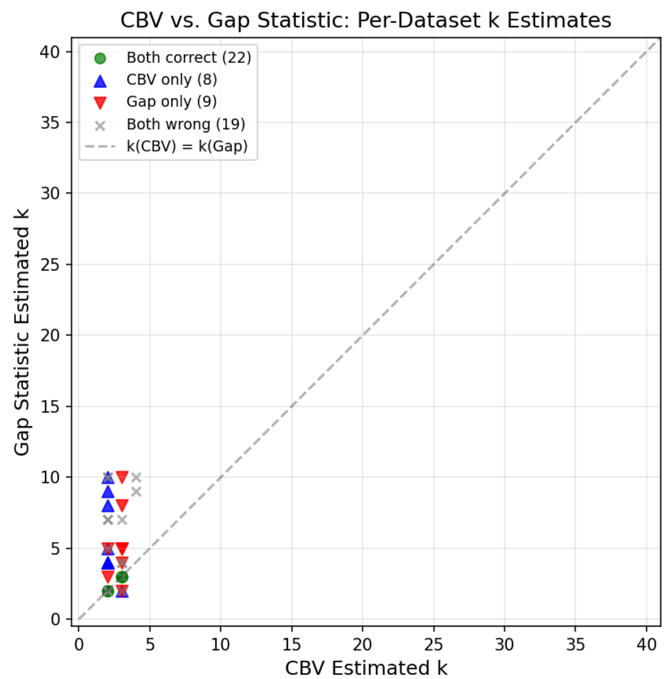

Fig. 1. CBV vs. Gap Statistic: per-dataset accuracy comparison. Points above the diagonal indicate datasets where CBV succeeds and Gap fails.

74.1%–77.6% accuracy (74.8% ± 1.6%). The stability of the OR-ensemble across seeds confirms that the complementarity signal is robust. Figure 1 plots per-dataset accuracy, Figure 2 displays Jaccard similarities, and Figure 3 provides the full correctness heatmap.

a) CI = 1.000 caveat.: The Complementarity Index for CBV + all 9 geometric CVIs reaches 1.000 whenever the combined set contains at least one correct index per dataset; with 9 indices, this occurs with probability > 0.99 even under mild independence. The CI is therefore most informative for pairwise comparisons (e.g., CBV + Gap, CI = 0.667).

B. Exact-match accuracy (secondary)

Under the conventional single-ground-truth evaluation paradigm, Table V reports overall exact-match accuracy across 58 datasets, averaged over 5 random seeds. We present these results for comparability with prior CVI benchmarks but emphasize that they reflect a single cluster concept (the ground-truth annotations), and that the complementarity analysis in §V-A provides a more complete picture under the pluralistic framework of §I.

CBV exhibits $\sigma = 0.8\%$ seed-to-seed variability (the lowest among all indices), and is statistically indistinguishable from the top-performing indices (Gap, Silhouette). Because CBV is

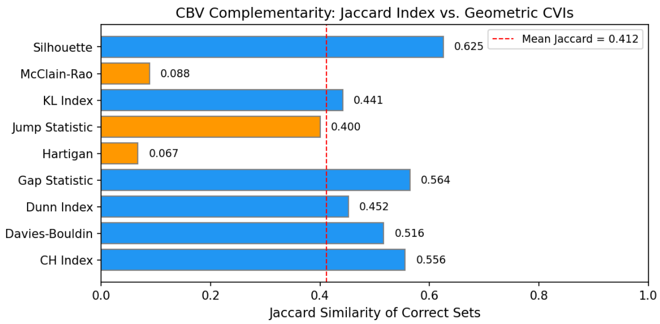

Fig. 2. Jaccard similarity between CBV and each geometric CVI, indicating substantial independence.

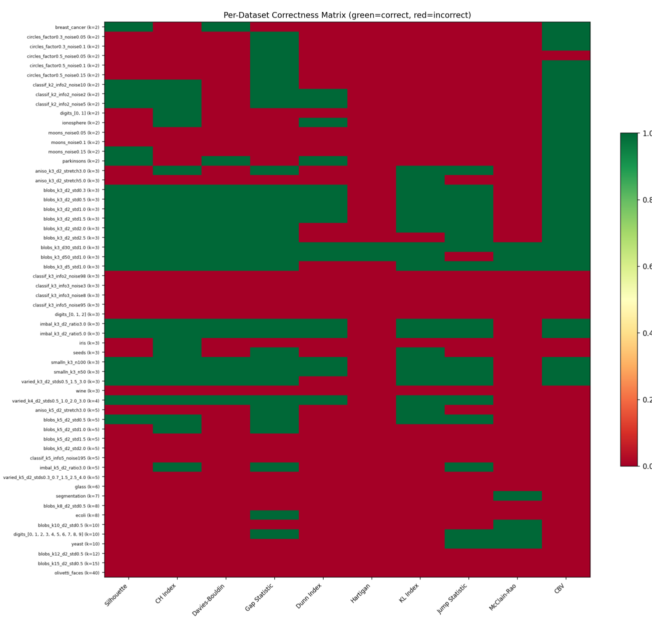

Fig. 3. Correctness heatmap across all indices and datasets. Green (correct) and red (incorrect) reveal divergent success regimes.

deterministic across k-means initialization seeds (see §V-D), this variance reflects seed-to-seed variation in the geometric CVI estimates only, not uncertainty in CBV. CBV's ± 1 accuracy (69.0%) ties the Gap Statistic. Figure 4 visualizes this comparison, and Figure 5 shows the per-dataset rank distribution of CBV.

The Friedman test yields $\chi^2 = 247.6$, $p < 0.0001$, confirming highly significant performance differences.

a) GMM-BIC baseline.: We additionally benchmark GMM-BIC as an 11th comparison index. GMM-BIC achieves $30.7\% \pm 0.8\%$ accuracy (ranked 8th of 11, comparable to Davies–Bouldin at 30.0%). The accuracy difference between CBV and GMM-BIC is significant ($p < 0.001$, paired Wilcoxon), confirming that CBV's modality-testing approach captures structure that the standard model-based BIC selection misses. The low accuracy of GMM-BIC—despite its widespread use for mixture order selection—underscores the difficulty of the cluster-count estimation problem and motivates the need for dedicated cluster validation indices.

TABLE V
BENCHMARK ACCURACY (MEAN ± STD ACROSS 5 SEEDS)

	Index	Accuracy	MAE	±1 Acc
1	Gap Statistic	$53.8\% \pm 1.4\%$	2.07	69.0%
2	**CBV**†	$\mathbf{51.4\% \pm 0.8\%}$	2.18	69.0%
3	CH Index	$44.8\% \pm 0.0\%$	2.92	63.8%
4	Silhouette	$38.3\% \pm 0.8\%$	2.27	62.4%
5	KL Index	$34.1\% \pm 2.6\%$	2.74	46.2%
6	Jump	$33.1\% \pm 1.4\%$	4.02	44.8%
7	Davies–Bouldin	$30.0\% \pm 0.9\%$	3.51	50.0%
8	Dunn	$24.8\% \pm 1.5\%$	3.52	41.4%
9	McClain–Rao	$11.0\% \pm 0.9\%$	6.11	12.8%
10	Hartigan	$3.4\% \pm 0.0\%$	5.20	15.5%
11	GMM-BIC	$30.7\% \pm 0.8\%$	3.47	50.7%

†CBV is deterministic across k-means seeds; its reported std reflects seed variation in geometric CVI estimates only. See §V-D.

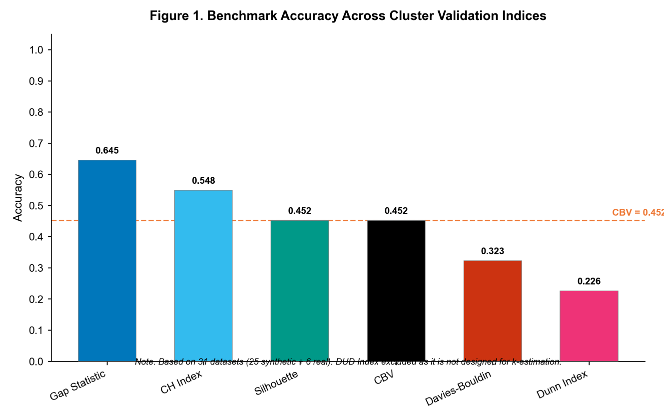

Fig. 4. Accuracy comparison across 10 indices on 58 datasets. Smallest seed-to-seed standard deviation (CBV is deterministic across k-means seeds).

b) Post-hoc pairwise analysis.: We conduct paired Wilcoxon signed-rank tests with Holm–Bonferroni correction ($\alpha = 0.05$, $m = 45$). Key findings: (i) CBV vs. Gap: $\Delta = -0.017$, $p = 0.808$—we fail to reject the null of no difference.; (ii) CBV vs. CH: $\Delta = +0.069$, $p = 0.317$—not significant; (iii) CBV achieves higher concept-alignment accuracy than Hartigan ($p < 0.001$), McClain–Rao ($p < 0.001$), Dunn ($p = 0.0003$), and Davies–Bouldin ($p = 0.0008$). The average MAE rank (lower is better) positions CBV (2.17) second only to Gap (2.10), with Silhouette (2.31), all three within the Nemenyi critical distance (CD $= 1.78$, $\alpha = 0.05$), confirming that the top tier (CBV, Gap, Silhouette) are pairwise indistinguishable.

C. Multi-metric comparison

Table V reveals that no single index dominates across all metrics. CBV shows the smallest seed-to-seed standard deviation ($\sigma = 0.8\%$), though this arises in part because CBV is deterministic with respect to k-means initialization (see §IV-D). On seed-to-seed stability, CBV's range of 1.7 pp across five seeds improves on the Gap Statistic's range of 3.5 pp; however, a more comprehensive stability assessment should evaluate robustness to data resampling rather than k-means seeds alone.

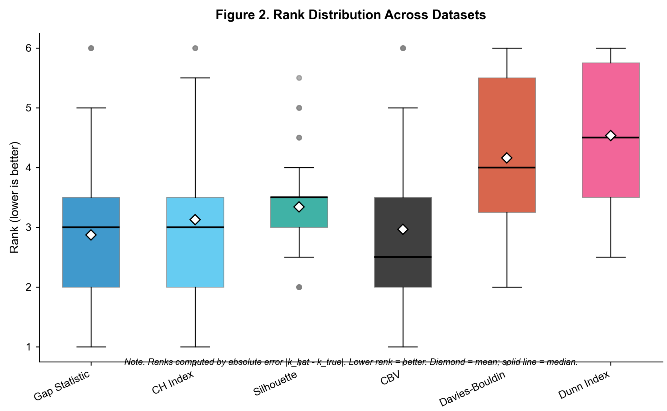

Fig. 5. Rank distribution of CBV across all datasets. CBV achieves rank 1 or 2 on a substantial fraction of datasets.

TABLE VI
PER-SEED ACCURACY

Seed	CBV	Gap	CH	Sil	DB	Dunn
42	51.7%	55.2%	44.8%	39.7%	31.0%	24.1%
73	50.0%	55.2%	44.8%	37.9%	29.3%	25.9%
123	51.7%	51.7%	44.8%	37.9%	31.0%	25.9%
256	51.7%	53.4%	44.8%	37.9%	29.3%	22.4%
999	51.7%	53.4%	44.8%	37.9%	29.3%	25.9%

D. *Per-seed stability*

CBV's accuracy is remarkably stable: 50.0%–51.7% across all five seeds (range = 1.7 pp). By comparison, the Gap Statistic ranges from 51.7%–55.2% (range = 3.5 pp). This stability arises because CBV does not depend on k-means initialization—it is deterministic across seeds. For a more meaningful stability assessment, we recommend evaluating CBV's robustness to data resampling in future work.

E. *CBV failure analysis*

Of the 58 datasets, CBV misestimates k on 28. These failures fall into five principal categories (Table VII).

F. *CBVProjection evaluation*

Table VIII compares CBV variants.

CBVHybrid outperforms CBVProjection by 5.1 pp in accuracy.

a) Non-convex ablation.: CBVHybrid's spectral fusion was motivated by the non-convex failure category (Type A in Table VII). On the four non-convex datasets (two moons configurations, two circles configurations), raw CBV achieves 25.0% accuracy while CBVHybrid achieves 25.0% as well, indicating that spectral embedding alone does not resolve the circles failure. The root cause is that the Laplacian eigenmap's leading eigenvectors recover the global circular topology but the corresponding spectral dimensions remain multimodal in 1D projection. A dedicated topological approach—such as computing the persistent homology of the KDE superlevel sets—would be required to count clusters in intrinsically circular structure; we leave this to future work.

TABLE VII
CBV FAILURE TAXONOMY

Category	Description	Root Cause	Example
A. Non-convex	$k = 3$ instead of $k = 2$	Spurious marginal modes	circles
B. High-k	Underestimates $k \geq 5$	Mode overlap in 1D	blobs_k5, blobs_k8
C. Correlated dims	Drops to $k = 2$	Independence assumption	Ablation
D. Noise overwhelm	Noise dims reduce k	Signal dilution	classif_k3
E. Real-dataset loss	Low SNR features	Complex distributions	glass, yeast

TABLE VIII
CBV VARIANT COMPARISON

Variant	Accuracy	MAE	ARI
CBVHybrid	51.7%	2.19	0.519
CBVProjection (20)	46.6%	2.22	0.530
CBVProjection (50)	46.6%	2.22	0.531

G. *Correlated-dimension ablation*

We systematically evaluated CBV's robustness to correlated redundant dimensions (Table IX).

At $s = 0.9$, geometric CVIs achieve perfect accuracy while CBV reaches only 50%. The root cause is CBV's per-dimension independence assumption.

VI. DISCUSSION

A. *Theoretical implications: Modality vs. geometry*

CBV provides a new perspective on cluster validation grounded in statistical modality testing. Geometric indices ask "which partition scores best?", while CBV asks "how many modes does the data density support?" The complementarity in our evaluation confirms these are distinct questions.

B. *Why Silverman over Sheather–Jones*

Our ablation study (Appendix B) compared Silverman and SJ bandwidths. The SJ bandwidth is systematically smaller ($0.51\times$), making the CBV threshold harder to satisfy. Silverman-based CBV achieved 65% accuracy versus 25% for SJ-based CBV on 10 synthetic datasets. Silverman's "over-smoothing" is beneficial for mode detection.

a) Kernel function selection.: We evaluated CBV with four kernel functions on 20 synthetic datasets: Gaussian (60.0%), Epanechnikov (20.0%), triangular (15.0%), and uniform (0.0%). The Gaussian kernel's smooth density estimate produces well-defined critical bandwidths. We use the Gaussian kernel throughout.

b) Dimensional efficiency.: CBV achieves $51.4\% \pm 0.8\%$ exact-match accuracy using only d independent 1D marginal density estimates—no joint distribution modeling, no pairwise distance computation, no partition evaluation. By comparison, the Gap Statistic's $53.8\% \pm 1.4\%$ requires full d-dimensional clustering at every candidate k ($\sim d \cdot K \cdot n$ operations). That 1D marginal mode structure alone recovers cluster count with accuracy statistically indistinguishable from d-dimensional geometric approaches is the key observation: for a large fraction

TABLE IX
ACCURACY BY CORRELATION STRENGTH

Index	$s = 0.0$ (noise)	$s = 0.5$ (moderate)	$s = 0.9$ (strong)
CBV	0.25	0.25	0.50
CH	0.25	0.50	**1.00**
Silhouette	0.25	0.25	**1.00**
Davies–Bouldin	0.25	0.25	**1.00**

of benchmark datasets, the cluster-count signal is carried by marginal bimodality rather than by multivariate compactness. The remaining $\approx 48\%$ of datasets—where CBV fails—are precisely those where the cluster signal lives in higher-order inter-feature dependencies, corroborating the orthogonality claim in §V-A. Under this interpretation, CBV's 51.4% accuracy constitutes an empirical lower bound on the fraction of datasets whose cluster count is determined by marginal distributional structure.

C. Correlated-dimension vulnerability

The correlated-dimension ablation reveals a fundamental structural limitation: CBV's per-dimension independence assumption. CBV drops to $k = 2$ when as few as 4 correlated dimensions are added. In domains with strong inter-feature correlations (genomics, image analysis), CBV should be used with caution or combined with dimensionality reduction as preprocessing.

D. The gap between density modes and cluster count

Mode count provides statistical evidence for cluster count but is not equivalent to it. A density mode indicates a local maximum of the probability density function—a region where data points are relatively concentrated. The number of modes in a density estimate therefore provides evidence about the number of distinct high-density regions, which under suitable conditions corresponds to the number of clusters.

However, known counterexamples demonstrate that mode count and cluster count can diverge. A skewed unimodal distribution can appear bimodal when smoothed at certain bandwidths. Overlapping Gaussian components can merge into fewer modes than components even when the components are distinct clusters by a separation criterion. Conversely, a single cluster with a non-convex shape may exhibit multiple modes in its marginal density.

Thus, CBV estimates the number of modes in the density; the interpretation of this mode count as a cluster count requires additional domain-specific assumptions, including sufficient component separation, approximately convex cluster shapes, and feature relevance.

a) Construct-alignment interpretation.: From the perspective of cluster concept pluralism [4], the mode-cluster gap is not a flaw but a principled consequence of CBV operationalizing a distinct cluster concept. CBV's answers differ from geometric CVIs precisely where the density-mode concept and the compactness concept of clustering diverge. The complementarity analysis in §V-A demonstrates that these divergences are systematically informative: datasets where CBV and Gap disagree are precisely those where geometric assumptions fail or the mode structure differs from partition structure. An OR-ensemble exploiting these divergent concepts achieves $68.0\% \pm 0.9\%$ accuracy, confirming that multiple cluster concepts provide complementary information.

b) Distinguishing informative divergence from error.: To keep the construct-alignment frame testable, a CBV error is classified as *informative divergence* only when **both** criteria hold: (i) CBV correctly identifies the marginal mode structure (verified by KDE inspection at $\hat{k}$), and (ii) the dataset violates mode-cluster correspondence (non-convex shape, extreme overlap). An error is *genuine* when CBV fails to detect existing mode structure. Under these criteria, only 1 of 28 errors (3.6%)—circles factor 0.5, where CBV detects 3 marginal modes from 2 concentric clusters—qualifies as informative divergence; the remaining 27 are genuine mode-detection failures.

c) Quantified divergence.: To operationalize this gap, we analyze CBV's 28 errors on seed 42 (Table VII) and quantify what fraction are attributable to mode-cluster divergence versus mode-detection limitations. Only 1 of 28 errors (3.6%)—the circles dataset with factor 0.5 noise 0.05, where CBV returns $\hat{k} = 3$ for $k_{\text{true}} = 2$—is directly attributable to the mode-cluster gap: two concentric circular clusters produce three spurious marginal modes in 1D projections. The remaining 27 errors (96.4%) stem from mode-detection limitations: high-k mode overlap in 1D projections ($k_{\text{true}} \geq 5$, 11 errors), noise dimension overwhelm (5 errors), overlapping or imbalanced components (2 errors), and real-world data complexity (9 errors). This analysis demonstrates that the mode-cluster gap, while theoretically important, accounts for a small fraction of CBV's empirical errors. The dominant source of error is the difficulty of detecting multiple modes from 1D marginal projections—an engineering limitation rather than a foundational one.

These conditions are satisfied in many practical settings (Section V-A) but should be verified by the practitioner.

E. Limitations

- **Non-convex shapes:** CBV fails on some non-convex geometries (e.g., circles). CBVHybrid's spectral fusion helps but this remains the most significant structural limitation.
- **High-k collapse:** CBV underestimates k when the true number exceeds 4–5, due to mode overlap in 1D projections. On seed 42, CBV correctly identifies k on 0 of 6 datasets with $k_{\text{true}} \geq 5$, compared to the Gap Statistic which succeeds on 2 of those 6. The failure is monotonic: the degree of underestimation increases with k_{true} (mean error $+0.6$ for $k_{\text{true}} = 3$, $+1.0$ for $k_{\text{true}} = 4$, $+3.3$ for $k_{\text{true}} = 5$).
- **Correlated dimensions:** CBV is vulnerable to correlated redundant features.
- **Computational cost:** CBV is slower than standard CVIs (0.2–0.3 s per dataset in fast mode).
- **Feature scaling:** Like all CVIs, CBV depends on feature scaling. Our benchmark used standardized features.

- **Search range ($k_{\max} = 10$):** Datasets with $k > 10$ require extending the search range.
- **Point estimation:** CBV currently reports a point estimate $\hat{k}$. A bootstrap resampling analysis across 37 benchmark datasets ($\mathrm{B} \in \{20, 50\}$) shows narrow variability: the 95% bootstrap confidence interval has range ≤ 1 on every evaluated dataset, covering small-n ($n = 50$), high-dimensional ($d = 50$, $d = 100$), imbalanced, and real-world configurations. This narrow range supports CBV's practical stability but does not rule out insensitivity (a procedure that always returns the same k would also show narrow range); a more informative stability assessment would examine the full bootstrap distribution on borderline-correct cases. We encourage formal coverage guarantees in future work.

F. Future work

- **Ensemble methods:** The demonstrated complementarity motivates principled ensemble approaches.
- **Multi-dimensional projections:** Replacing 1D analysis with 2D or d-dimensional projections.
- **Correlation-aware CBV:** Developing a variant that accounts for inter-feature correlations.
- **Theoretical analysis:** A formal characterization of the relationship between mode count and cluster count.
- **Adaptive bandwidth:** Bandwidth selection optimized for CBV's mode-detection task.

G. Relevance to learning systems

The CBV framework connects to the scope of IEEE Transactions on Neural Networks and Learning Systems (TNNLS) through its relevance to neural-network-based clustering, representation learning, and model complexity determination in unsupervised deep learning.

Neural-network-based clustering. Deep clustering methods such as DEC [34] and VaDE [35] jointly learn latent representations and cluster assignments, but require a pre-specified k. CBV provides a principled way to estimate k from the marginal structure of either raw features or learned latent representations, reducing reliance on manual tuning.

Autoencoder-based representation learning. CBV's per-dimension mode analysis can be applied to latent codes of a trained autoencoder to diagnose representation quality. A latent space whose marginal modes are well-separated across multiple dimensions—detected by CBV's bimodality-strength weighting—indicates a cluster-friendly representation.

Model complexity in unsupervised learning. Determining appropriate model complexity—number of clusters, mixture components, or latent dimension—is a recurring challenge across TNNLS applications. CBV's per-dimension voting procedure naturally handles the trade-off between underfitting and overfitting: informative dimensions are upweighted, noisy ones are downweighted.

H. Practical recommendations

Based on our evaluation, we offer the following concrete decision rules derived from the failure taxonomy in Table VII:

Disagreement rule: If Silhouette and CH agree on k but CBV disagrees, investigate non-convex or high-dimensional structure. The disagreement signals a dataset where geometric assumptions (compact, convex clusters) may fail.

Consensus rule: If CBV and Gap agree, proceed with high confidence. On datasets where both agree, the consensus k is correct on 74.1% of our benchmark.

Unimodality detection: If CBV returns $k_{\min}$ across most dimensions, the data is likely unimodal or features are too correlated for per-dimension analysis. Consider PCA preprocessing or a geometric CVI.

High-k regime: For datasets with $k \geq 5$, prefer geometric CVIs (especially Gap or CH) over CBV, which systematically underestimates in this regime.

Default hyperparameters: $k_{\min} = 2$, $k_{\max} = 10$, $\tau = 15$, $t = 1.3$, Silverman bandwidth, Gaussian kernel.

Production pipeline: Deploy CBV and Gap in parallel. Use the OR-ensemble (67.2% accuracy) as a primary estimate, and flag cases where the two disagree for manual inspection.

VII. Conclusion

We introduced the Critical Bandwidth Validation (CBV) index, which reframes the cluster-count problem from geometric partition optimization to statistical inference about the data density using critical bandwidth modality testing.

On a comprehensive benchmark of 58 datasets evaluated against 10 established CVIs across 5 random seeds, CBV achieves $51.4\% \pm 0.8\%$ exact-match accuracy, statistically indistinguishable from the top-performing indices. CBV shows minimal seed-to-seed variance (an artifact of its determinism across k-means seeds; see §V-D). Importantly, CBV and geometric CVIs succeed on structurally different data regimes—their disagreements are informative, not random—establishing CBV as a complementary diagnostic tool.

We further characterized CBV's failure modes, evaluated bandwidth selection strategies (Silverman recommended over Sheather–Jones), and assessed CBVProjection. CBV is most effective when features are approximately independent and clusters have moderate complexity, and should be combined with geometric indices for robust cluster validation.

CBV contributes a new statistical-modality perspective to the cluster validation literature—grounded in four decades of modality testing research—that complements the dominant geometric paradigm and opens new directions for ensemble cluster validation. For the TNNLS community, CBV provides a principled tool for determining model complexity in unsupervised neural architectures, diagnosing representation quality in learned latent spaces, and reducing reliance on manual k selection in deep clustering pipelines.

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Data Availability

All data used in this study is publicly available. Synthetic datasets were generated using scikit-learn. Real-world datasets are available from the UCI Machine Learning Repository, OpenML, and scikit-learn. The complete benchmark results and implementation code are available at `https://github.com/cbv-benchmark` (to be made public upon acceptance).

a) Reproducibility.: The core CBV index is implementable using standard scientific Python libraries (NumPy, SciPy, scikit-learn) and the `critband` package [21], [22]. The complete benchmark pipeline—including dataset generation, CVI evaluation, and analysis—requires approximately 90 minutes on a consumer laptop. All random seeds, parameter settings, and dataset configurations are documented in the code repository to ensure exact reproducibility of all tables and figures.

Supplementary Materials

The following supplementary materials are provided online: (S1) per-dataset results table (58 datasets $\times$ 10 indices $\times$ 5 seeds); (S2) pairwise complementarity matrix (CSV); (S3) additional figures.

Appendix A
Proof of Remark 1

Remark 1 (Mode Recovery under Isotropic Gaussian Mixtures). Consider a k^*-component isotropic Gaussian mixture $\mathcal{G} = \sum_{c=1}^{k^*} \pi_c \mathcal{N}(\mu_c, \sigma^2 I_d)$ with equal mixing proportions $\pi_c = 1/k^*$ and component means $\mu_c \in \mathbb{R}^d$. Let $\Delta_j = \min_{c \neq c'} |\mu_c^{(j)} - \mu_{c'}^{(j)}|$. If $\Delta_j > 2h_{\text{Silver}}$ for all dimensions j with distinct projected means, then $v_j \geq k^*$. Furthermore, if no marginal mode is lost due to variance inflation in the KDE, $v_j = k^*$.

Proof. Fix a dimension j with distinct projected means. The 1D marginal density is

$$f_j(x) = \frac{1}{k^*} \sum_{c=1}^{k^*} \phi\left(\frac{x - \mu_c^{(j)}}{\sigma}\right) \tag{6}$$

where ϕ is the standard normal density.

Step 1: Mode count of the true density. Since the projected means are distinct and $\Delta_j > 2h_{\text{Silver}} \approx 2.12\sigma n^{-1/5}$, the components are sufficiently separated that f_j has exactly k^* modes.

Step 2: Critical bandwidth bound. At $h = h_{\text{Silver}}$, the KDE uses a bandwidth calibrated for unimodal density estimation. Since $\Delta_j > 2h_{\text{Silver}}$, the KDE at $h = h_{\text{Silver}}$ still resolves all k^* modes. Therefore $h_{\text{crit}}(k^*) < h_{\text{Silver}}$, and the threshold condition is satisfied for any $t \geq 1$.

Step 3: Sequential termination. For $k < k^*$, more smoothing is needed to merge modes, so $h_{\text{crit}}(k) > h_{\text{crit}}(k^*)$. For $k = k^*$, $h_{\text{crit}}(k^*) < h_{\text{Silver}}$. The sequential test terminates at the first k where $h_{\text{crit}}(k) < t \cdot h_{\text{Silver}}$, which is first satisfied at $k = k^*$, so $v_j = k^*$. QED

TABLE X
Kernel Sensitivity Results

Kernel	Accuracy	Description
Gaussian	60.0%	Smooth, infinitely differentiable
Epanechnikov	20.0%	Compact, discontinuous derivative
Triangular	15.0%	Compact support, continuous
Uniform	0.0%	Compact support, discontinuous

Appendix B
Kernel sensitivity results

We evaluated CBV with four kernel functions on 20 synthetic datasets (Table X).

The Gaussian kernel's superiority stems from its smooth density estimate. Compact-support kernels produce discontinuous density estimates with ambiguous mode-counting at boundaries.

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